Milaan

Closing the books before noon

Payment reconciliation for the analyst who has four files and one close to finish.

The Singleton · Hemanth Chilakala

ExaThon India 2026 · Track 1

Meet the analyst

> A reconciliation analyst at a mid-sized Indian NBFC can't close the day's books on time because UPI, NACH and payment-gateway settlement files arrive in different formats with mismatched references, so they chase breaks by hand in Excel. If this worked, they'd close the books before noon and spend their time only on the exceptions that actually matter.

Milaan reconciles synthetic receipts for the fictional Sahyog Finance Ltd,
using Exasol Personal for normalisation, matching, scoring and audit records.

One credit. 37 payments.

37 PayFast payments

Gross ₹6,07,085.27

− fees ₹4,856.70 − GST ₹874.21

Bank credit ₹6,01,354.36

36 ledger receipts present · 1 missing

Loan LN-100000-36 · missing receipt ₹16,425.71

Why spreadsheets fail

References disagree
Truncated bank narrations and different receipt identifiers.

Amounts disagree
Gross EMIs, net settlements, fees and GST.

Dates disagree
Delayed credits, holidays and next-day NACH returns.

Illustrative consequences: suspense balances, wrong customer follow-ups and slower audit review.

Milaan in one picture

Ledger · Bank · Gateway · NACH

↓ bulk IMPORT

EXASOL PERSONAL

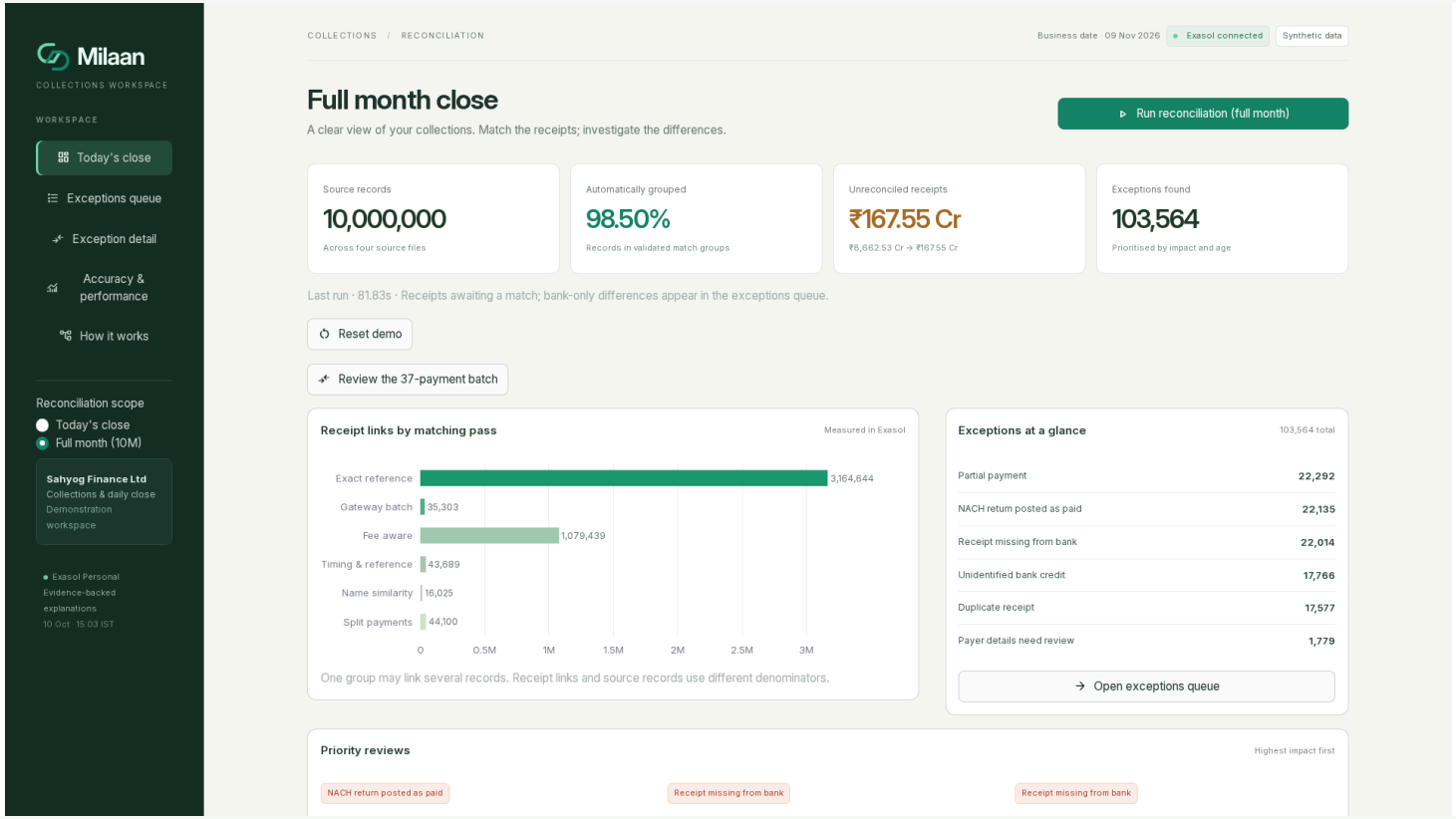
SQL normalisation → seven passes + UDFs → ranked exceptions

↓ small result sets

Analyst review → decision audit in Exasol

Separate ground truth → SQL scoring

Watch the close progress



Full month for scale · Today's close for arriving records and carried open items

Seven passes. One shrinking queue.

EXACT	**3,164,644**	3,164,644 cumulative
PG_BATCH	**35,303**	3,199,947 cumulative
FEE_AWARE	**1,079,439**	4,279,386 cumulative
TOLERANT	**43,689**	4,323,075 cumulative
FUZZY	**16,025**	4,339,100 cumulative
SPLIT_MERGE	**44,100**	4,383,200 cumulative
CLASSIFY	**0**	4,383,200 cumulative

Measured 10M run · counts are receipt links · CLASSIFY labels remaining exceptions

Why Exasol

76.14 seconds

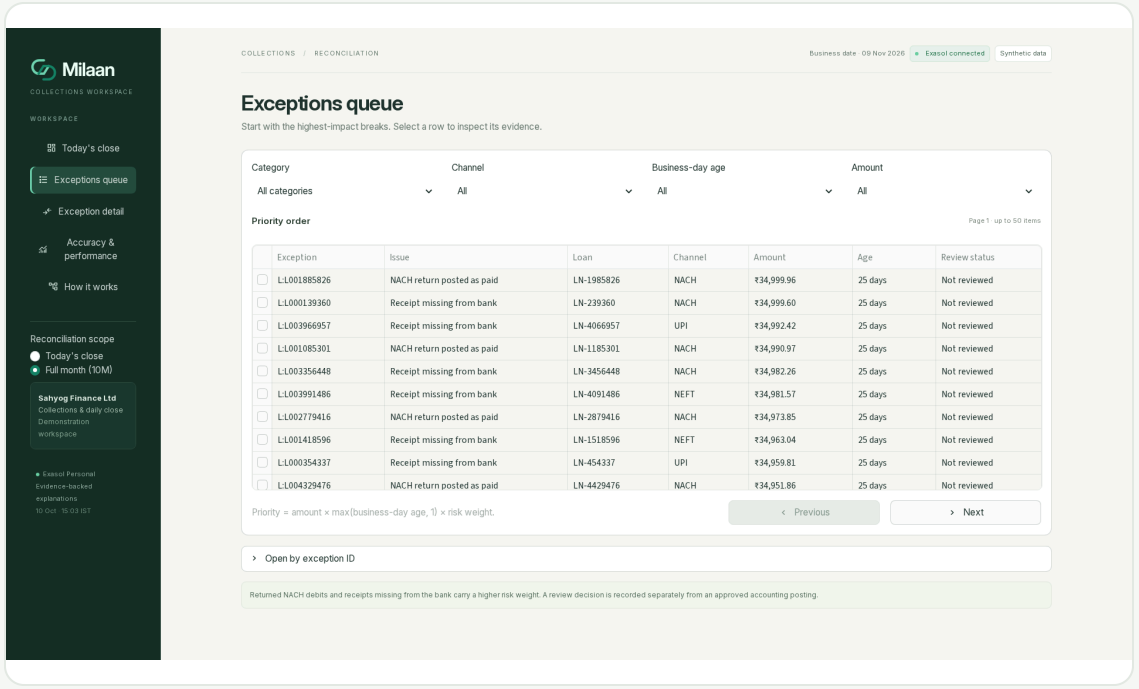
10,000,000 source records · 4 cores · 15.62 GiB RAM

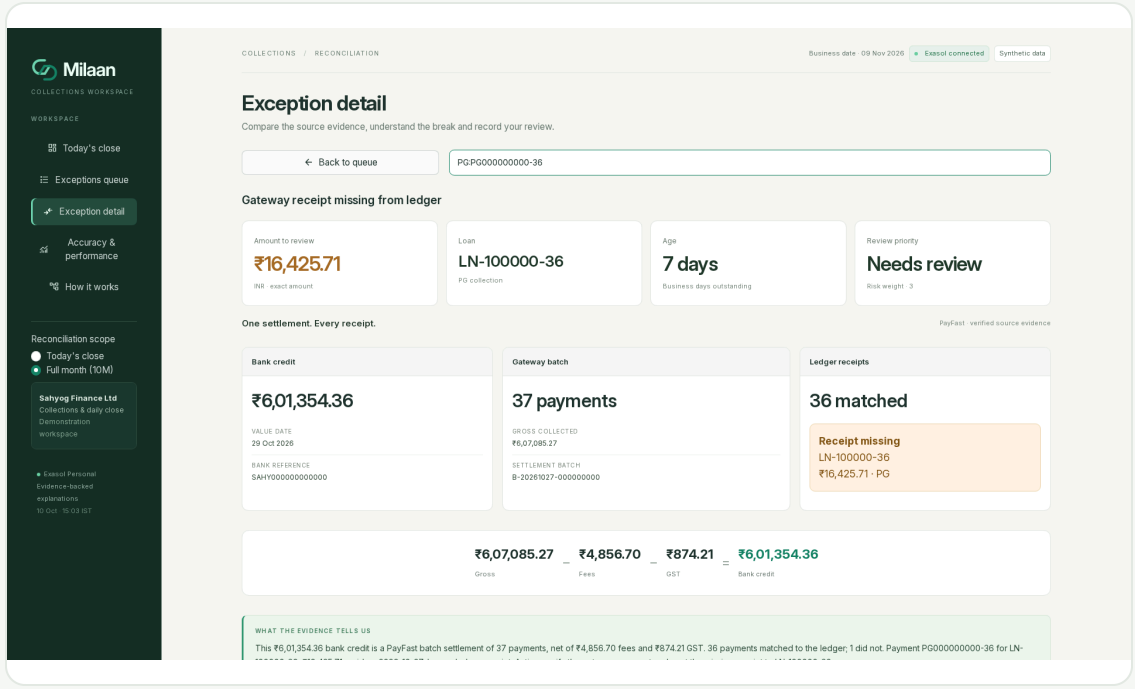
Bulk IMPORT · hash joins · REGEXP · windows · Python scalar · Lua SET UDFs

EXACT: 19.27s · PG_BATCH: 6.04s · FEE_AWARE: 7.15s · TOLERANT: 1.56s · FUZZY: 1.13s · SPLIT_MERGE: 3.04s · CLASSIFY: 12.77s

Database-side results, validation and scoring included. Explanations and profiling flush follow the timer.

Review the money that matters





Prioritise → inspect evidence → accept, reject or assign

Every review records the analyst, timestamp and reasoning in Exasol.

Accuracy, tested three ways

1M dataset	Precision	Recall	False links
Normal	100.000%	100.000%	0
Blind	100.000%	100.000%	0
Hard	99.154%	85.872%	3,220

10M normal: 100.000% precision · 99.959% recall · 98.502% source membership

Blind: unseen seed and break rates. Hard: collisions, reversals, longer delays and extra splits.

Trust comes from evidence

Deterministic explanations
Amounts, dates and rules come from stored evidence. No external inference service.

Human review
The analyst accepts a recommendation, rejects it or assigns an owner.

Auditable decisions
Run-linked evidence and review history remain in Exasol.

A final membership check rejects conflicting match groups.

A practical path to impact

First: independently labelled validation with an NBFC partner.

Next: live feeds, real holiday calendars, authenticated review roles and approved ledger write-back.

Then: measure larger datasets and concurrent operations.

Similar reconciliation workflows exist in banks, insurers, fintechs and gateway-based merchants.

Milaan

Close with confidence

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https://github.com/Hemanth-Chilakala/milaan

Demo video: docs/video/demo.mp4

MIT · All data is synthetic

Appendix: limitations & methodology

- Seeded, controlled data; production accuracy needs independent partner labels.
- Frozen normal/blind/hard rules; bank reversals explain hard-mode false links.
- Split search: at most 12 candidates, subsets of two or three.
- Single node and one worker; no real ledger connector or review authentication.
- Timing includes preparation, results, validation and scoring; browser/recording load changes it.
- Daily/monthly equivalence tested on 1,000 records; full 10M daily replay remains unmeasured.
- Codespace rebuild and clean checkout tested; macOS/WSL remain unverified.

Run IDs, profiles and raw observations: docs/benchmarks.md